Course Project: Full-Stack Web Application

CMP 357: Web Application Programming

Project Overview

This project requires you to work in teams of **3–4 students** to design and implement a full-stack web application.

Each team must **select one of the three project ideas** described below and implement all its required features. In addition, each team must design and implement **at least two additional functional features** of their own choice.

Students are expected to develop their own application design, including the layout, styling, navigation, and user experience.

The system will consist of:

- **Front-End:** HTML, CSS, JavaScript, and Bootstrap for styling and interactivity. The interface must be **responsive and adapt to different screen sizes**, e.g. desktop and mobile devices.
- **Back-End:** Node.js and Express for handling requests, implementing application logic, and generating dynamic content.
- **Database:** MySQL for storing, managing, and retrieving application data.

Project Ideas

Each team must choose **one** of the following three projects.

Project 1: Medical Expert Q&A Platform

Description:

Develop a web application that connects users with medical experts from different specialties. Doctors can register and create professional profiles. Users can browse experts, select a suitable doctor, and submit questions either publicly or privately. Public questions and their answers can be accessed by other users.

Required Features:

- User registration and login for regular users and medical experts.
- Expert profiles displaying specialties and qualifications with an average rating of their previous responses.
- Users can browse and search for experts using different attributes and submit questions to the selected expert.
- Users can choose whether their questions are public or private.
- Experts can view and respond to questions assigned to them.
- Users can browse and search previously answered public questions.
- Users and experts can track or delete their questions and responses through personalized dashboards.
- Users can rate the responses of experts.

Project 2: Professional Mentorship Platform

Description:

Develop a web application that connects individuals seeking career or educational guidance with experienced professionals. Mentors can create profiles highlighting their expertise and offer mentorship sessions. Users can discover suitable mentors, request and book sessions, and exchange advice.

Required Features:

- User registration and login for mentors and mentees.
- Mentor profiles displaying professional experience, expertise, and available time slots.
- Mentors can create and publish posts on their profiles, including text, images, and videos, to promote their mentorship services.
- Users can search for mentors by specialization or professional field.
- Users can request mentorship sessions, which mentors can accept or reject.
- The system should prevent overlapping bookings.
- Mentors and mentees can exchange messages or questions related to their sessions.
- Users can view their upcoming and previous mentorship sessions and provide feedback.

Project 3: Pet Care Bidding Platform

Description:

Develop a web application that connects pet owners with individuals offering pet-care services, such as pet sitting, walking, and feeding. Pet owners can post requests describing the services they need, while care providers compete by submitting bids. Pet owners can compare offers and select the most suitable provider.

Required Features:

- User registration and login for pet owners and care providers.
- Pet owners can create pet profiles including details and photos of their pets and post care requests specifying the required service, dates, and budget with a bidding deadline.
- Care providers can browse available requests and submit bids with proposed prices and service descriptions.
- Care providers should be able to search and filter requests based on type of the pet, the required service, budget, etc.
- Providers can edit or withdraw their bids before the bidding deadline.
- Pet owners can review submitted bids and select a winning provider.
- The system should manage request and bid statuses, preventing new bids after a request is closed.
- Users can track their active and completed requests and leave reviews after completed services.

Project Requirements

All teams must complete the following components:

(1) Project Team Formation

Fill in the following form with the team names and selected project idea.

(2) Implementation

- Implement all required features of the selected project.
- Design and implement **at least two additional functional features** of your own choice. These must provide meaningful functionality beyond the required features.
- Develop an original, user-friendly, and responsive front-end using HTML, CSS, JavaScript, and Bootstrap. The application must work properly on different screen sizes.

- Implement a Node.js and Express back-end to handle requests, responses, and application logic.
- Use MySQL for persistent data storage, with appropriate database relationships.
- Include appropriate client-side and server-side input validation.

(3) Intermediate Project Milestone

- Each team is required to submit a working demo of their front-end implementation by **Wednesday, 4 November 2026**.
- The front-end demo should use sample data to demonstrate the application's functionality without requiring an actual back-end implementation.

(4) Final Deliverables and Report

- A working demo of the complete web application with a readme file to explain how to navigate the web app.
- A 3-minute video showing the application and its main features.
- A report (5–6 pages) including:
  - Description of the application, implemented features, and use cases.
  - Description of the two additional features designed by the team.
  - Explanation of back-end processing, implementation details, and database schema.
  - Clear description of the contribution of each team member.

Notes

- Teams are free to choose their own application design, layout, styling, and additional functionality.
- Although the project is completed in teams, each student will be graded individually.
- Individual grading will be based primarily on a one-on-one discussion with each student about the project's code and implementation. During the discussion, students will be asked to explain specific parts of the code, the application logic, database interactions, and design decisions. Students may also be asked to trace the execution of a feature or explain how a modification would be implemented. Students who fail to demonstrate sufficient understanding of their code may lose the entire project grade.
- Academic honesty is expected. Any external code or resources used must be properly acknowledged.